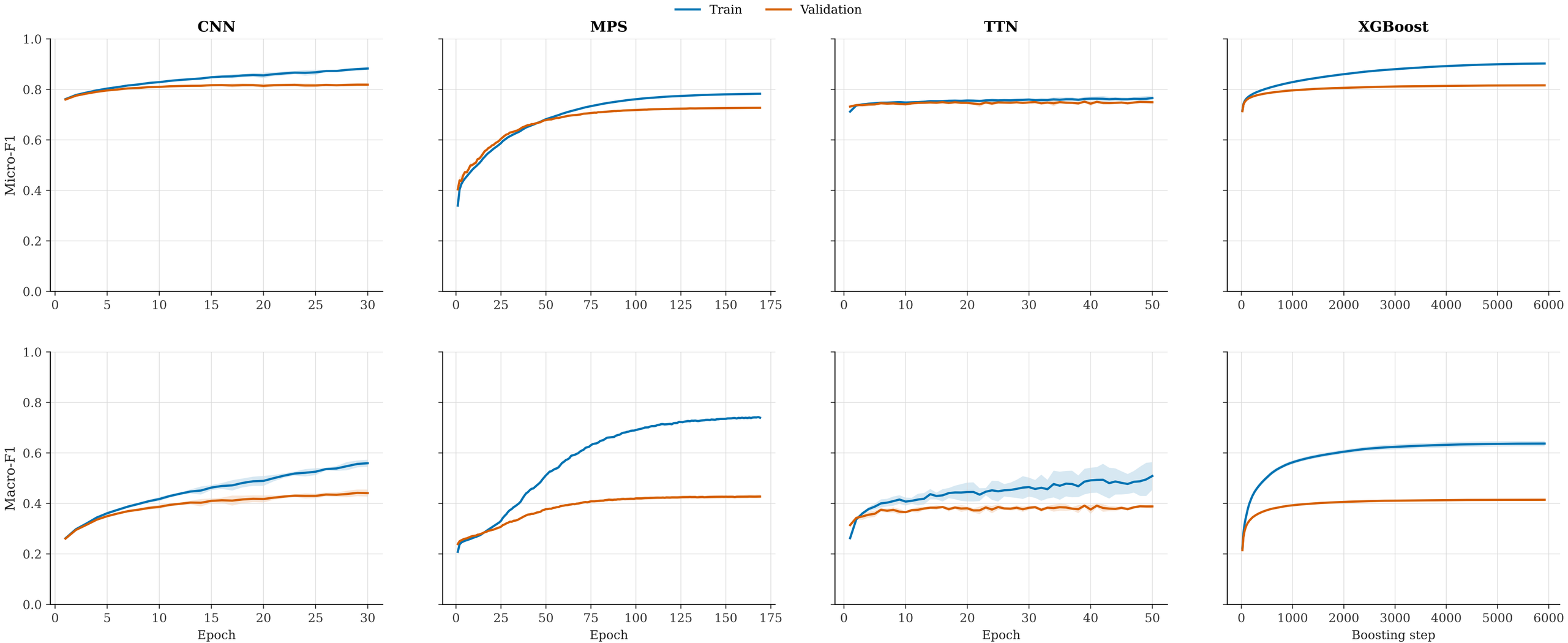
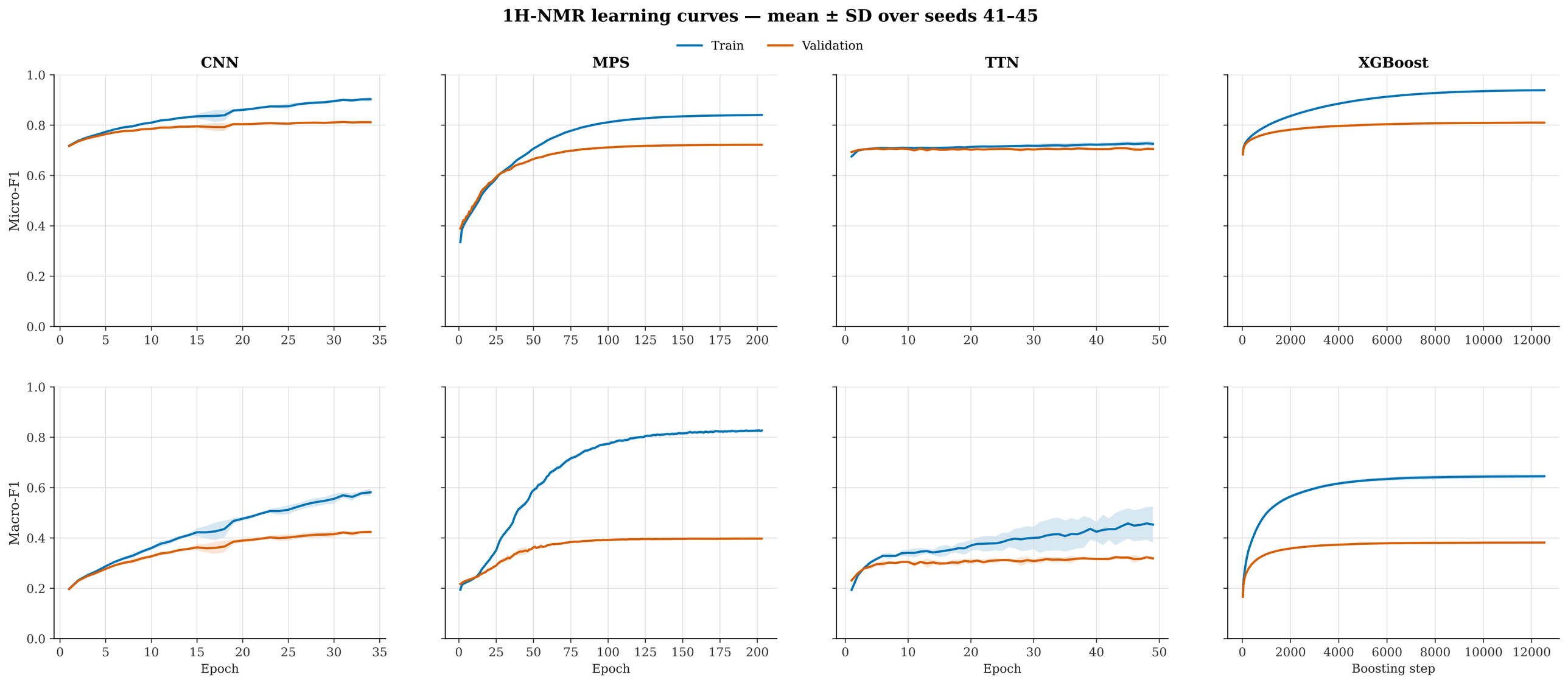
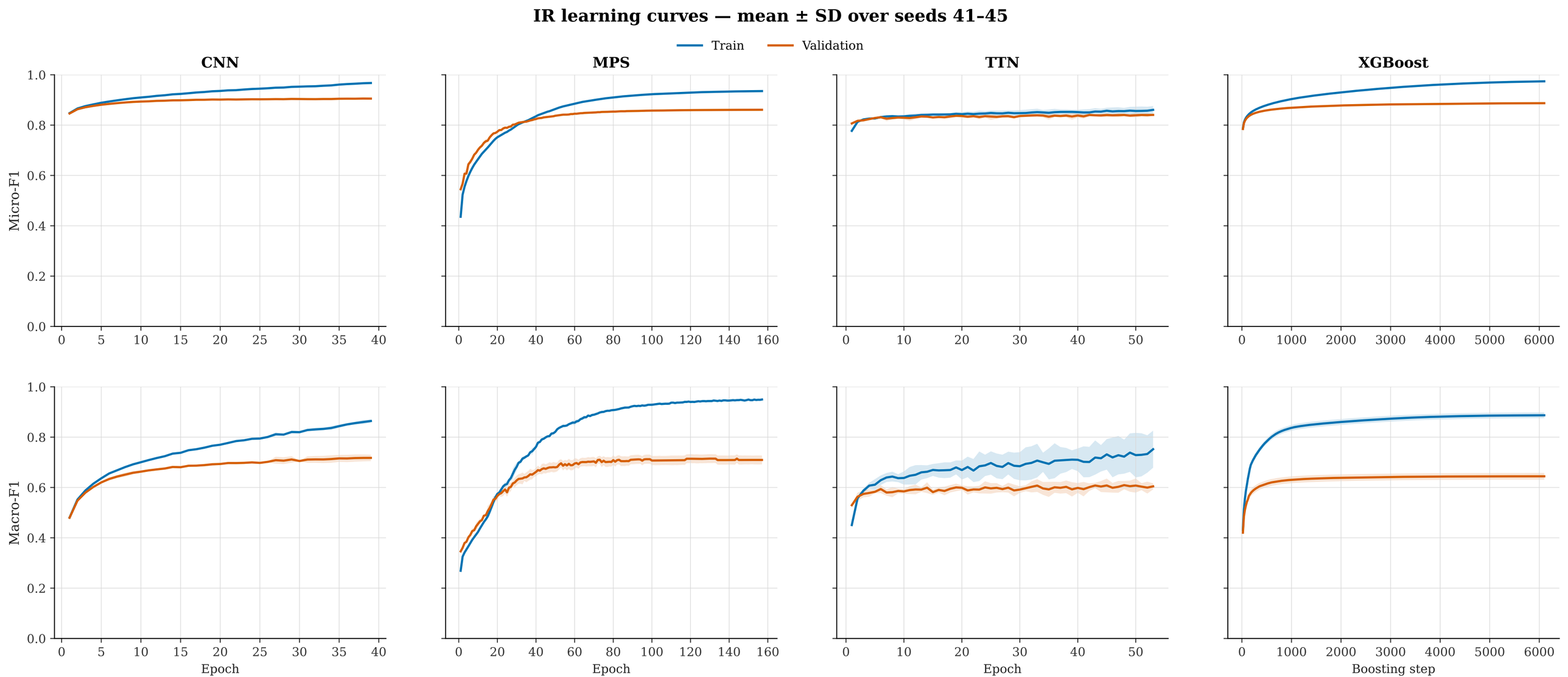
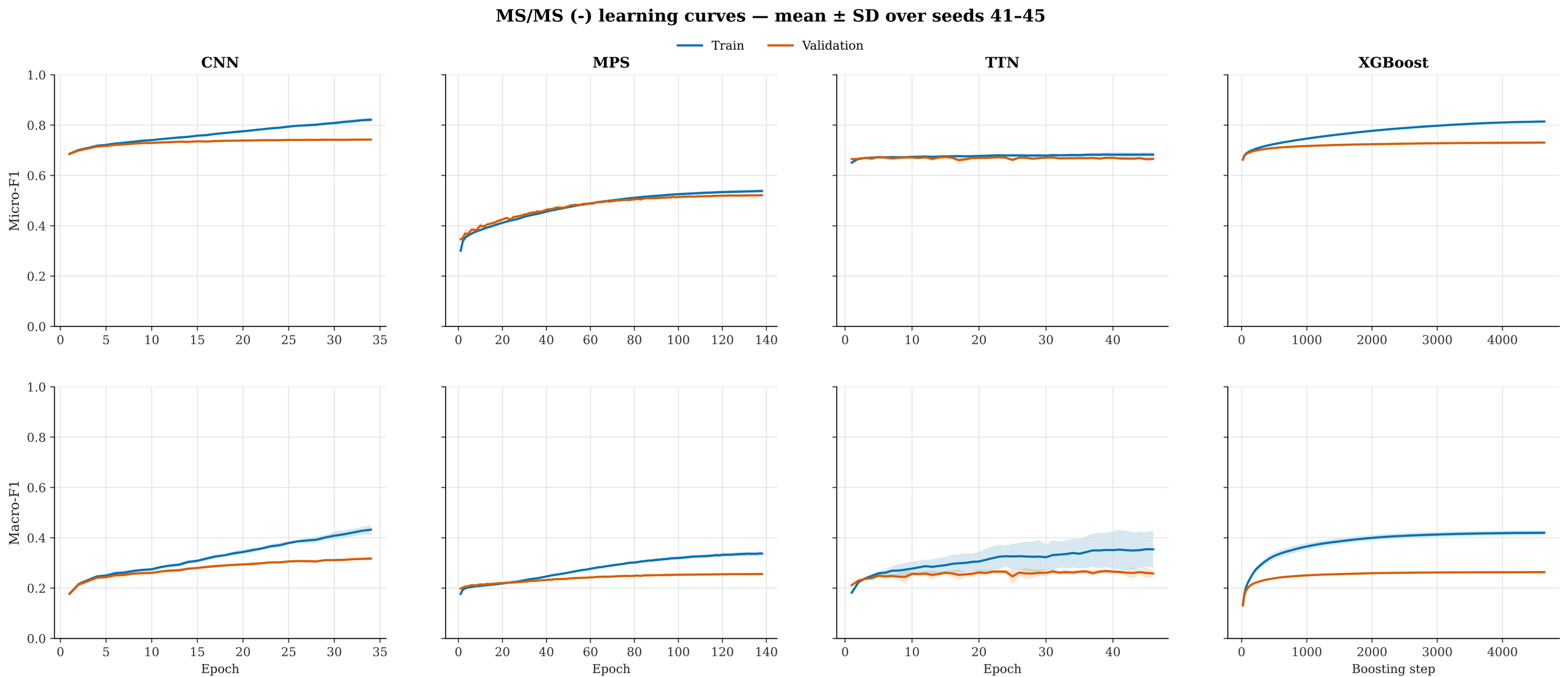
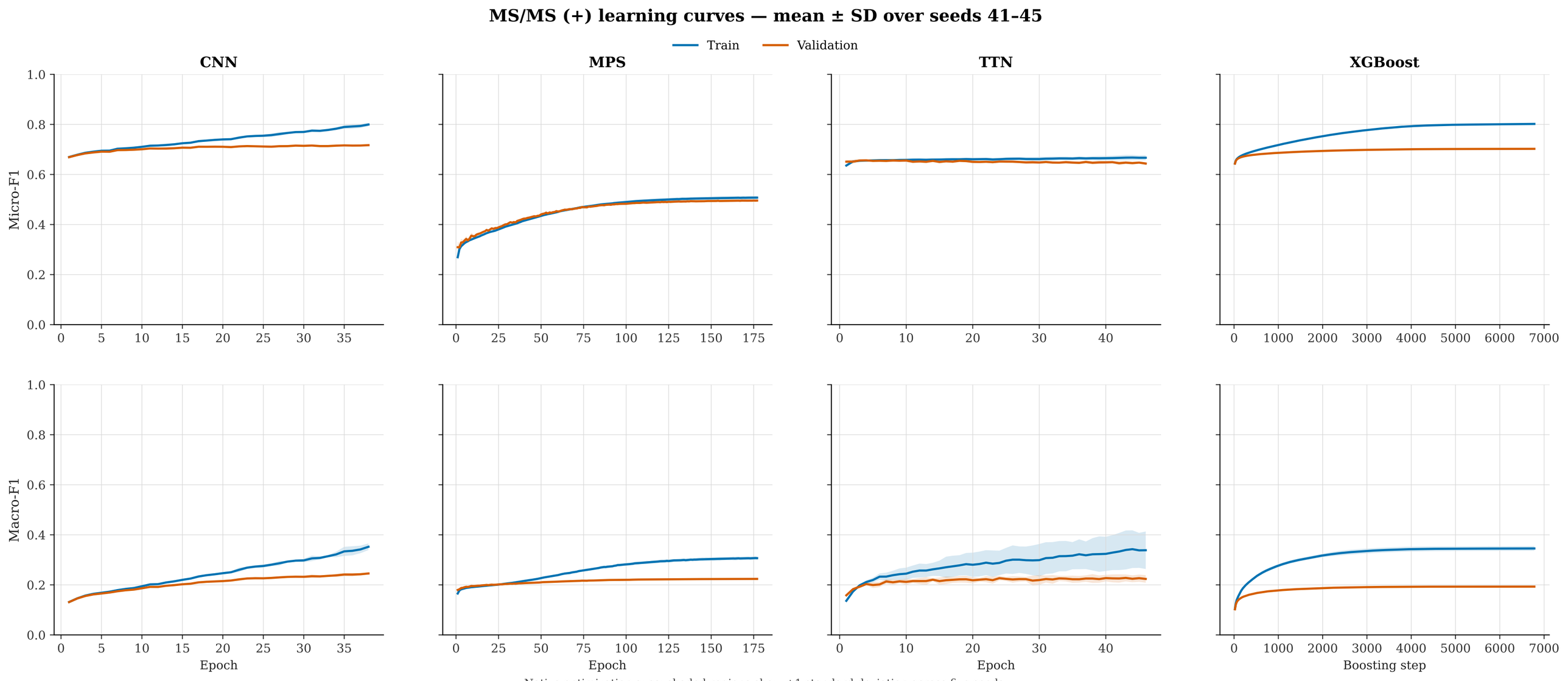

13C-NMR learning curves — mean ± SD over seeds 41–45
Train
Validation
CNN
MPS
TTN
XGBoost
Micro-F1
Macro-F1
Epoch
Epoch
Epoch
Boosting step
Native optimization axes; shaded regions show ±1 standard deviation across five seeds.


1H-NMR learning curves — mean ± SD over seeds 41–45
Train
Validation
CNN
MPS
TTN
XGBoost
Micro-F1
Macro-F1
Epoch
Epoch
Epoch
Boosting step
Native optimization axes; shaded regions show ±1 standard deviation across five seeds.


IR learning curves — mean ± SD over seeds 41–45
Train
Validation
CNN
MPS
TTN
XGBoost
Micro-F1
Macro-F1
Epoch
Epoch
Epoch
Boosting step
Native optimization axes; shaded regions show ±1 standard deviation across five seeds.


MS/MS (-) learning curves — mean ± SD over seeds 41–45
Train
Validation
CNN
MPS
TTN
XGBoost
Micro-F1
Macro-F1
Epoch
Epoch
Epoch
Boosting step
Native optimization axes; shaded regions show ±1 standard deviation across five seeds.


MS/MS (+) learning curves — mean ± SD over seeds 41–45
Train
Validation
CNN
MPS
TTN
XGBoost
Micro-F1
Macro-F1
Epoch
Epoch
Epoch
Boosting step
Native optimization axes; shaded regions show ±1 standard deviation across five seeds.